

Uncertainty-aware Decision Support in Power Grid Demand Forecasting

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Abstract

Effective management of modern power grids requires accurate demand forecasting and interpretable decision support for fluctuations in electricity consumption. This study extends the Calibrated Explanations method with online calibration support for uncertainty-aware probabilistic regression in offline, semi-online, and fully online settings. Since the application is a temporal forecasting problem, the proposed workflow should be understood as an empirically updated online decision-support procedure rather than as a new finite-sample conformal-validity theorem for arbitrary dependent time series. We apply Long Short-Term Memory (LSTM) networks and online ridge regression to an electricity-demand dataset to provide calibrated forecasts with quantified uncertainty. These forecasts provide threshold-risk information and support operator interpretation through Calibrated Explanations, including sensitivity analysis of key variables that affect energy demand. The proposed approach illustrates the practical use of online calibrated explanations in power-grid decision support.

Keywords: Uncertainty-aware decision support, Online explanations, Electricity demand forecasting, Power grid management, Calibrated Explanations, Probabilistic regression, Conformal predictive systems

1. Introduction

Power-grid operators need short-term demand forecasts that identify threshold risk and explain which conditions affect that risk. This paper addresses that need through explainable probabilistic decision support in a real-world business case involving Jönköping Energi, a Swedish energy company. Jönköping Energi operates in the local and regional energy markets, providing energy production, distribution, and communication services primarily to customers in and around the Jönköping municipality. The company’s subsidiary, Jönköping Energi Nät AB, manages the local power grid and will face challenges due to increasing electricity consumption and the integration of renewable energy sources (Pinheiro et al., 2023).

To address these challenges, Jönköping Energi has initiated the exploratory project *Local Flexibility*, which aims to establish a local market mechanism for flexibility in energy usage. This project encourages high-consumption customers to reduce their electricity usage during peak demand periods, a process known as peak-shaving (Martins et al., 2018). Accurate demand forecasting is crucial for the success of this initiative, as it enables informed decision-making and optimizes the utilization of the power grid.

The power grid is monitored continuously for anomalies, power outages, electrical power quality, and more. In work on *Local Flexibility*, decisions must be made in a data-driven manner and supported by tools that explain suggested actions. Rather than providing only single-point estimates, the models used in this paper output prediction intervals for power consumption. A defined threshold, related to the power-subscription limit, is used to evaluate operational risk.

Operators are mainly interested in a forecast horizon of 12–48 hours, because this period gives both the operator and the customer time to act if needed. The business model is still under development, and this study is part of a broader exploratory analysis before a local flexibility market is deployed. Hourly demand forecasts over 12–48 hours are operationally relevant for such decisions. At longer horizons, hourly forecasts are treated as less reliable for customer-level decisions, which can affect customers’ daily operations. To keep the evaluation simple and interpretable, this paper uses a fixed 24-hour forecast horizon. Future work will consider more complex, near-real-world decision settings.

The main contribution of this paper is an online extension of Calibrated Explanations, an existing local explanation method that both calibrates and explains predictive models. The extension supports offline models with online calibration, semi-online models with occasional updates, and fully online models with continuous learning and calibration. It also adds user-defined alternatives, enabling decision makers to explore threshold-based “what-if” conditions at prediction time. Together, these additions provide threshold-risk information and local explanations in an online decision-support workflow. We do not claim a new conformal-validity result for dependent time-series data, instead the contribution is practical and methodological. Classical conformal guarantees rely on assumptions such as exchangeability, whereas the present power-demand setting is temporal and may exhibit autocorrelation, seasonality, and drift.

The remainder of this paper is structured as follows: Section 2 presents the techniques used, Section 3 describes the contribution, Section 4 describes the experimental setup, Section 5 presents the results, and Section 6 provides conclusions and future work.

2. Theoretical Background

This study presents an online explanation method for pre-trained models used with online calibration, semi-online models with occasional updates, and fully online models that are updated and calibrated as new instances become available. Consequently, two predictive models are used: a pre-trained Long Short-Term Memory (LSTM) network and a lightweight online ridge regression model.

The LSTM network uses historical data to train a model offline. In contrast, the online learning approach, exemplified by the ridge regression model, allows incremental updates as new data become available, making it more adaptable to evolving patterns. This dual-

method strategy enables an examination of the trade-offs between offline and online learning paradigms. Both scenarios use online calibration, where new data update the calibration over time.

2.1. LSTM

An LSTM is a type of Recurrent Neural Network (RNN) capable of learning both long-term and short-term features of training data (Muzaffar and Afshari, 2019). This type of neural network addresses time-series problems while mitigating the vanishing-gradient problem that can occur in RNNs when modeling long-term dependencies (Sherstinsky, 2019). Compared to a standard RNN cell, an LSTM cell includes a specialized vector known as the internal cell state, which acts as long-term memory storage. Additionally, three gates are present in an LSTM cell: the input gate, the forget gate, and the output gate. The input gate controls the flow of new data into the memory unit, the forget gate determines how much historical data is discarded, and the output gate specifies the final output data (Hao et al., 2023). Information within the LSTM cell is processed and filtered using the sigmoid (σ) and hyperbolic tangent ($\tanh$) activation functions (Wang et al., 2022).

This architectural design enables LSTMs to filter irrelevant data, preserve important information, and mitigate the vanishing-gradient problem, making LSTM networks well suited for time-series problems.

2.2. Online Ridge Regression

Ridge regression, and its special case, least squares, are classical regression algorithms. Least squares goes back to Gauss and Legendre, while modified ridge regression was suggested in the 1960s. In matrix form, ridge regression can be represented as

$$\omega_n = (X_n^T X_n + aI_p)^{-1} X_n^T Y_n, \quad (1)$$

where $X_n = (x_1, \dots, x_n)^T$ is the $n \times p$ data matrix, and $Y_n = (y_1, \dots, y_n)^T$ is the vector of corresponding labels. The scalar $a \geq 0$ is the ridge parameter, and the case $a = 0$ corresponds to ordinary least squares. For a test object x , the ridge regression prediction is

$$\hat{y} := x^T \omega_n. \quad (2)$$

The matrix $H_n := X_n(X_n^T X_n + aI_p)^{-1} X_n^T$ is sometimes called the *hat matrix*, since it is used to transform y_i to $\hat{y}_i$ using

$$\hat{Y}_n := (\hat{y}_1, \dots, \hat{y}_n) = H_n Y_n. \quad (3)$$

It is worth noting that the matrix $(X_n^T X_n + aI_p)^{-1}$ can be updated efficiently online as new examples arrive using the Sherman–Morrison formula, see Bartlett (1951), avoiding a costly matrix inversion at each step.

Furthermore, ridge regression is fully online, making it possible to train and calibrate using the same data.

2.3. Calibrated Explanations

Calibrated Explanations (Löfström et al., 2025; Löfström et al., 2024) is a recently proposed explanation method with the following core characteristics:

- Calibration of model predictions to provide accurate estimates of both regression outputs and class probabilities.
- Uncertainty quantification of calibrated predictions, indicating the degree of uncertainty in the prediction.
- Feature rules conveying feature importance with uncertainty quantification.
- Exploration of alternative predictions, making it possible to construct counterfactual, semifactual, and superfactual explanations with uncertainty quantification.
- Conjunctive rules, indicating joint feature contributions with uncertainty quantification.
- Thresholded predictions for regression, providing the probability of the target being below (or above) a given threshold. This is a unique explainability feature of Calibrated Explanations.

The method relies on a calibration model outputting a point prediction with an uncertainty interval. The calibration model is fitted for the test object being predicted and explained. Once calibrated, it is used on perturbed variations of the test object to explore feature rules that define feature importance or alternative predictions. The method systematically perturbs the test object, one feature at a time, to explore the calibration-model output for instances similar to the original test object. Once outputs have been collected for the perturbed instances, two different kinds of rules can be formed:

- Factual explanations composed of feature rules with a rule condition and uncertainty-aware feature importance, or
- Alternative explanations with a rule condition indicating what uncertainty-aware prediction a changed test object would result in.

The calibration method used for regression is Conformal Predictive Systems (Vovk et al., 2019). More broadly, conformal prediction provides distribution-free uncertainty quantification under exchangeability assumptions (Vovk et al., 2022; Angelopoulos and Bates, 2021). Venn-Abers (Vovk and Petej, 2014) is used as the calibration method for the probabilities. Conformal Predictive Systems can provide the probability of the target being above or below a given threshold, which is calibrated using Venn-Abers to obtain the uncertainty interval. Related conformal regression methods include conformalized quantile regression, which combines conformal calibration with quantile regression to obtain intervals that adapt to heteroscedasticity (Romano et al., 2019). For a more detailed description of Calibrated Explanations for regression, including Conformal Predictive Systems and Venn-Abers, the reader is referred to Löfström et al. (2025).

3. Contribution

3.1. Calibrated Explanations for Time Series

Calibrated Explanations in its original form assumes an offline scenario, where a trained model is used to explain each test object. Time-series forecasting, however, can be performed online or semi-online, with models updated continuously or at regular intervals. Regardless of whether an offline, semi-online, or fully online setup is used, test instances can be added to the calibration set once the target value becomes available. This makes it possible to update the calibration continuously as new test objects are predicted and explained. To facilitate calibrated predictions and explanations for time series, the Calibrated Explanations framework is extended to support offline, semi-online, and fully online prediction and explanation with online calibration updates.

Classical conformal validity arguments rely on exchangeability, an assumption that is often strained in time-series settings where autocorrelation and distribution shifts are expected. Adaptive and online conformal-inference methods address such settings by updating calibration under distribution shift (Gibbs and Candès, 2021, 2022), and conformal time-series methods such as EnbPI provide related tools for constructing intervals under temporal dependence (Xu and Xie, 2021). This paper focuses on the explanation layer. It extends Calibrated Explanations so that calibrated threshold probabilities, uncertainty intervals, and local explanations can be updated and used in online decision support.

Accordingly, the validity status of the proposed workflow must be interpreted carefully. When the assumptions of the underlying calibration procedure hold, the usual conformal or probabilistic calibration interpretation applies. In the present time-series application, however, we treat the reported threshold probabilities and uncertainty intervals as empirically calibrated, chronologically updated uncertainty estimates for decision support. We do not claim finite-sample distribution-free validity for arbitrary dependent sequences. This distinction is important because the aim of the paper is to extend the explanation and calibration workflow for online use, not to replace dedicated adaptive or time-series conformal methods with explicit online-coverage guarantees.

In this application, the daily cycle also motivates hour-aware online calibration since observations from the same clock hour are more comparable than arbitrary neighboring hours, which is why the online ridge-regression setup updates the model and calibration in an hour-specific manner.

There are three scenarios to consider when adding support for time series in Calibrated Explanations. If online learning is used, the solution must support continuous learning and calibration, whereas an offline model requires continuous or batch calibration. For semi-online learning, occasional model updates must also be supported.

3.1.1. FULLY ONLINE SCENARIO:

In the online scenario, the model is partially fitted when the target for a previous test object is revealed; this can be done either for each instance or for a batch of instances. Any model update may change predictions on the previously used calibration set, making it necessary to reinitialize the calibration model after some updates. The calibration procedure takes this into account and reinitializes the calibration model only when predictions have changed. In other words, calibration scores are not treated as fixed if the underlying model update

changes the predictions on which those scores depend. In that case, the calibration model must be rebuilt, or the affected scores must be recomputed, before the calibration output is used again. The efficient insertion of a new residual into the sorted residual list is used only when the predictions for the existing calibration objects are unchanged. For regression, when no changes appear for past data, the residual for the added instance is inserted into the sorted list of residuals used by the Conformal Predictive System. Thus, new instances can be calibrated efficiently for regression¹.

This implementation rule is intended to avoid stale calibration scores after online model updates. It should not be read as a proof of exchangeability-based validity under temporal dependence.

3.1.2. OFFLINE MODEL WITH ONLINE CALIBRATION SCENARIO:

In the offline scenario, the model is used without retraining. Here, *offline* refers to the fixed predictive model, not to a static calibrator. Updating the calibration set with newly revealed outcomes allows the empirical uncertainty estimates to reflect recent operating conditions without retraining the predictor. An important difference compared with the online scenario is that the same data cannot be used for both training and calibration. One solution is to grow the calibration set continuously using only previously seen test instances; this means that a number of test instances must be encountered before a sufficient calibration set has been collected. An alternative is to set aside part of the training set to initialize a sufficiently large calibration set, and then use previously seen test instances to grow it as additional targets become available. In this scenario, the predictions for the calibration set do not change, making it possible to update the Conformal Predictive System efficiently by adding only the residuals for the new calibration instance(s).

In this scenario, model training and calibration remain separated. New test objects are added to the calibration set only after their target values have become available. Thus, at prediction time, the calibration set contains only past revealed outcomes and does not include the target currently being predicted.

3.1.3. SEMI-ONLINE SCENARIO:

The semi-online scenario is similar to the *offline model/online calibration* scenario, except that the model can be updated occasionally. This scenario requires strict handling of training and calibration data so that the two sets are not mixed. This handling is assumed to be managed by the user.

3.2. User-defined Alternatives

In addition to the support for time-series data, another feature is added to support decision making. The rule conditions proposed by Calibrated Explanations are automatically identified and may not correspond to the conditions that decision makers want to investigate. To allow decision makers to explore different conditions, custom rules are implemented. This can be particularly useful when demand is high and weather forecasts are uncertain. While

1. As this paper focuses on regression, efficient updates for classification are left as future work. The version used in this paper reinitializes the Venn-Abers calibrator for each classification update.

new rules may be added to both factual and alternative explanations, they often provide the most value when exploring alternatives, since they provide insight into what can be expected under user-defined conditions. Once a rule is added, it can be combined with rules covering other features to explore joint feature contributions under various conditions.

4. Experimental Setup

4.1. Data

All experimental data are available at hourly resolution. Energy consumption is extracted from internal systems at Jönköping Energi, while weather parameters, such as temperature and wind speed, originate from the Swedish Meteorological and Hydrological Institute (SMHI). The consumption series represents aggregated power load from underlying sub-stations, consumers, and producers. As shown in Table 1, three years of hourly data, covering 2021–2024, are used for training and evaluation. Before use, the data are validated for anomalies such as missing values, duplicates, and erroneous values. The overall data quality is high, with few anomalies.

Variable	Source	Unit	Resolution	Data points
Historic consumption	Internal measuring	MW	Hour	26 137
Air temperature	SMHI	°C	Hour	26 137
Wind speed	SMHI	m/s	Hour	26 137
Working day	Calendar	True/False	Hour	26 137

The training procedure differs between the two algorithms. The LSTM is trained of-line using separate training and validation sets, divided approximately into 2021–2022 for training and 2023 for validation. Ridge regression is pre-trained using the first three years and updated online during the test period.

The full test period consists of data from January to April 2024. The first 350 test instances have already been revealed and used for calibration before the period visualized in the results is evaluated.

The initial training set for the ridge regression models consists of data collected over the last three years for the same season as the test set. The test set is then processed in the online setting, expanding the training set each time a new measurement becomes available.

4.2. LSTM

For the LSTM architecture, data are normalized with min–max scaling. The LSTM network also uses a fixed input length during both training and inference. Therefore, a sliding window of 168 time steps, corresponding to one week, is used. The model includes a temperature-forecast variable, which is important for demand forecasting.

During model training, the LSTM network can be configured with several hyperparameter settings, each leading to different performance. Hyperparameter tuning is performed by grid search over settings such as the number of LSTM cells, the number of LSTM layers, and the batch size. The evaluated hyperparameters and their ranges are shown in Table 2,

together with the chosen settings for the final *champion* model. The Adam optimizer is used because of its well-known performance in LSTM networks (Hewamalage et al., 2021; Kingma and Ba, 2017). Furthermore, a learning-rate scheduler is introduced during training, lowering the learning rate every fifth epoch. Mean Absolute Percentage Error (MAPE) is used only to select the point-forecasting model during the hyperparameter grid search.

Table 2: Hyperparameters in the LSTM tuning process

Hyperparameter	Range	Champion model
LSTM layers	1-2	1
LSTM cells per layer	25-200	50
Dropout	0.0-0.4	0.0
Batch size	1-500	8

4.3. Online Ridge Regression

The ridge regressor and the online calibrated explainer are trained and calibrated on all data prior to the test period. Each object x_t consists of the demand at time t , the clock hour, whether it is a working day, and the temperature forecast for time $t + 24$. Once the true demand is observed 24 hours later, the true temperature is also known, so the ridge regressor is updated online using the true temperature in place of the forecasted temperature. Each object x_t is used to predict demand 24 hours ahead, and y_t denotes the corresponding target value. Consequently, the target associated with x_t becomes available 24 hours later. At time t , the ridge regressor is therefore updated by adding (x_{t-24}, y_{t-24}) to the training set. The temperature forecast in x_{t-24} is replaced by the corresponding observed temperature before the update, and the ridge parameter is tuned using generalized cross-validation (Golub et al., 1979).

4.4. Calibration

The LSTM is trained offline, and only calibration is performed online. For both modelling approaches, calibration is chronological, meaning that an observation can contribute to calibration only after the corresponding target has been revealed. Thus, the online calibration set available at a prediction time contains previous revealed outcomes, not the target value currently being forecast. Ridge regression is continuously fitted and calibrated in a fully online fashion. Since the business case is tied to operational consequences when a specific threshold is exceeded, we use probabilistic regression. Throughout the results, reported probabilities are probabilities of remaining below the threshold; exceedance risk is the complement. The data are normalized to protect confidential business information, and the normalized value 0.86 corresponds to the true operational demand threshold used in the case study. The `calibrated-explanation` package is used for experimentation².

2. Calibrated Explanations with support for adaptive behavior can be made available upon request once the paper is published. Data cannot be shared due to company restrictions.

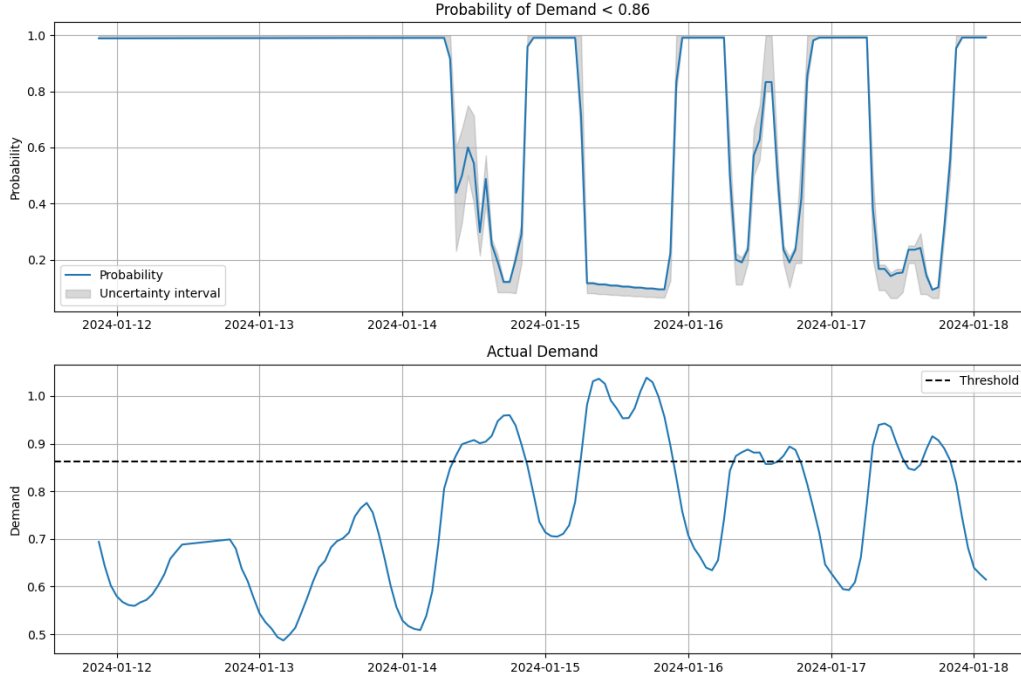


Figure 1: Top: Predicted probabilities of remaining below the critical threshold for the LSTM model. Bottom: Demand with the threshold marked as a dotted line.

5. Results

The results are divided into two parts, predicted probabilities over the visualized period and example explanations. The figures should be read as illustrative examples of the on-line decision-support output rather than as a complete empirical validation of conformal coverage or probability calibration. A full operational validation would additionally require aggregate reliability, threshold-event, and efficiency metrics over the complete evaluation period. One reason for using Calibrated Explanations in a regression context is that it provides uncertainty-aware probabilities for the target being below (or within) a user-defined threshold (or lower and upper bound thresholds), which can be determined by the user at prediction time.

Figure 1 shows the probabilities from the LSTM model (top, blue), with uncertainty shown in gray. The true demand and the threshold are shown below. The model indicates when threshold risk changes over time. As can be seen, the probabilities mostly reflect true threshold exceedance with elevated uncertainty bands close to the threshold crossing.

Figure 2 shows the probabilities, with uncertainty, for the same period using ridge regression. The ridge regression model uses only four core features and a simple linear form, but still illustrates how a lightweight online model can provide threshold-risk information. As can be seen, the uncertainty is generally very high whenever the demand is expected to increase. This is an effect of both the fact that ridge regression is a weaker linear model and the fact that each clock-hour-specific calibrator receives one new observation per day.

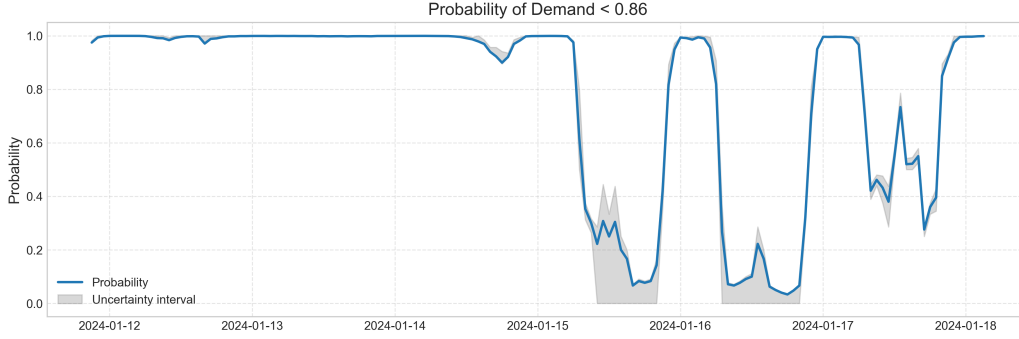


Figure 2: Predicted probabilities of remaining below the critical threshold for ridge regression.

Comparing LSTM and ridge regression, it is evident that the stronger model is better at detecting some of the threshold crossings. The first crossing, on the 14th of January, is detected by the LSTM but missed almost entirely by the ridge regression model. Since the LSTM has the moving window included among its input features, it has a much stronger signal to work with. The ridge regression model uses only its limited feature set when predicting, having a greater difficulty detecting the first crossing event in this time series.

Temperature is used in the explanation examples because it is a practically important driver of demand in this case study and because operators can meaningfully reason about alternative temperature forecasts.

The predicted probabilities with uncertainty indicate when operators may need additional interpretation. If the system indicates critical demand, it is useful to examine which input conditions contribute to that assessment. Since temperature forecasts are part of the model input and are important for electricity demand, operators can explore how the probability changes if the temperature increases or decreases. Figure 3 illustrates a factual explanation for the LSTM model from January 14, with a manually added rule for the temperature forecast at prediction time, i.e., 24 hours ahead. The feature *-0 Temperature 24* uses -0 to indicate the prediction time, i.e., zero hours before the prediction time. The two bars at the top illustrate the probabilities of being below or above the threshold, with uncertainty shown in lighter blue or red. The feature rules are independent of each other and contain inclusive conditions on the left. Red rules indicate fulfilled conditions that contribute toward the probability of exceeding the threshold. Blue rules instead contribute toward the probability of remaining below the threshold. The lighter areas indicate the degree of uncertainty in the feature importance. The top rule is created by the algorithm and shows that the probability of high demand can mainly be attributed to the temperature forecast being below 0.31.

A manually added rule was introduced to show how the prediction is affected when the forecast is above a lower temperature value (0.1 in this case, in blue). The rule shows that, because the temperature forecast is above 0.1, the probability of lower demand is non-negligible. In other words, the lower forecast bound helps explain why the estimated exceedance risk is not higher, which is because it pulls the prediction toward lower demand.

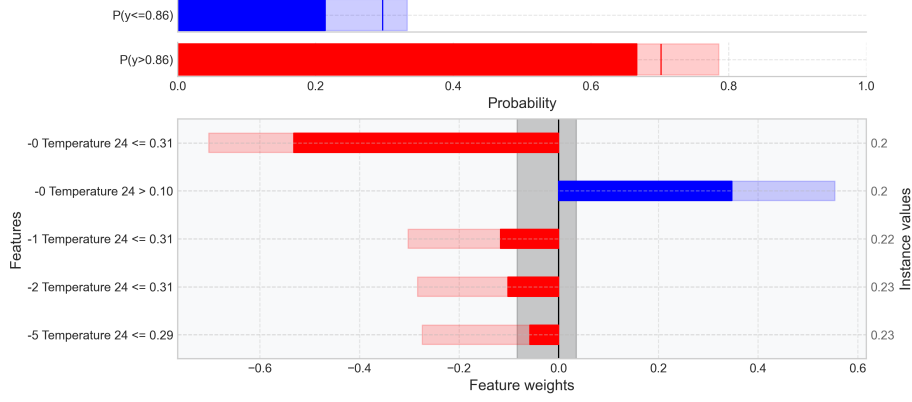


Figure 3: A factual LSTM explanation with a manually added rule for higher temperature.

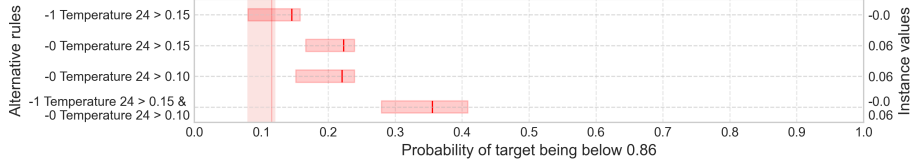


Figure 4: An alternative LSTM explanation with a manually added rule.

It is also possible to explore what predictions to expect under alternative inputs. Figure 4 illustrates an alternative explanation from January 15, with manually added rules and conjunctive rules that indicate the joint contribution of two rules. The light area covering the full height shows the probability for the true test object, whereas the alternative rules indicate how the prediction would change under other temperature forecasts.

The alternative explanation shows that higher temperature forecasts at prediction time reduce the predicted risk of exceeding the threshold, while also increasing uncertainty for some perturbations. This illustrates how user-defined alternatives can expose both the direction of a decision-relevant effect and the uncertainty associated with exploring conditions that differ from the original forecast.

6. Conclusions

We extend Calibrated Explanations for online use and illustrate the resulting uncertainty-aware decision-support workflow using real-world power-grid demand data. The method supports offline models with online calibration, semi-online models with occasional updates, and fully online models with continuous learning and calibration. The current contribution is practical and exploratory: it demonstrates how calibrated threshold probabilities, uncertainty intervals, and local explanations can be incorporated into an online decision-support workflow for power-grid demand forecasting. Standard regression is also fully supported, as is classification, although the implementation is primarily optimized for efficient online calibration in regression.

6.1. Future work

Controlled synthetic time-series experiments with configurable seasonality, autocorrelation, drift, and threshold-event frequency would enable more systematic stress testing of calibration behavior and explanation stability. Future evaluations should also report calibration-update and explanation latency across calibration-set sizes and model-update frequencies, including the cost of recomputing scores after predictor updates.

A natural next step is a full quantitative evaluation of empirical calibration and decision utility over longer operational periods, including reliability curves for threshold probabilities, threshold-event detection metrics, and uncertainty-width or interval-efficiency measures.

The proposed solution provides insight into features in the object used for prediction. Thus, solutions using sliding windows to capture temporal aspects are straightforward. Online models that inherently incorporate temporal aspects will provide insight into the core features, without temporal highlights. Furthermore, optimization of online calibration for classification is left as future work.

It is natural to use the probabilities and their uncertainty as signals for operator alerts. Defining an operator alert interface is within the industrial scope for putting the system to use. Future research should also incorporate richer weather inputs, preferably with forecast uncertainty, as temperature is perceived to be a strong indicator of energy consumption.

Future work should also consider filters for implausible perturbations, since some user-defined alternatives may combine feature values in ways that are unlikely under the training and calibration distributions.

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